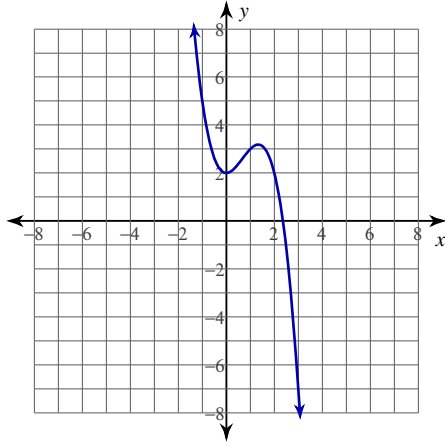


Intervals of Increase and Decrease

Date _____ Period _____

For each problem, find the x-coordinates of all critical points, find all discontinuities, and find the open intervals where the function is increasing and decreasing.

1) $y = -x^3 + 2x^2 + 2$



2) $y = x^3 - 11x^2 + 39x - 47$

3) $y = -x^4 + 3x^2 - 3$

4) $y = \frac{x^2}{4x + 4}$